The PB-ROE Valuation Model Revisited

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ABSTRACT

A two-stage stock valuation model derived from Wilcox's (1984) *P*/*B-ROE* approach is shown to be a surprisingly effective tool for a broad variety of uses, including the explanation of current prices and the prediction of future return differences. This applies both to the cross-section of individual US stocks and the time-series of the S&P 500 index. In addition to offering a new closed-form expression for firm value, the model measures the investment horizon over which the market predicts exceptional profitability, demonstrates the dependence of expectations for future profitability on past volatility of return on equity, and quantifies the extent of the recent US stock price bubble. Perhaps its most persuasive result is its no-look-ahead statistical support for tactical asset allocation.

I. INTRODUCTION

A good model relating stock prices to fundamental variables is of use to:

1. Corporate managers who want to understand how best to increase the value of their firm,

2. Fundamental analysts who wish to evaluate corporate managements and predict the results of their efforts,

3. Buyers and sellers who need to set prices for risky assets not already well-priced by liquid markets, and

4. Active investors who attempt to forecast abnormal returns based on mismatches between current price and indicated equilibrium prices supported by a firm's fundamentals and current macroeconomic conditions.

Clearly, no single model will be absolutely best for all applications. A model optimized as an explanatory tool for corporate managers may work poorly as a predictive tool for active investors, and vice versa. Likewise, a parsimonious linear approximation may be more usefully estimated, and more useful for predicting stock returns, than a more complex and comprehensive model that provides precise insights.

Surprisingly, however, one can accomplish competitive results in several directions with a relatively simple and robust two-stage valuation model structure built on Wilcox's (1984) *P*/*B*-*ROE* approach. It is our purpose in this paper to enhance the *P/B-ROE* model, and to illuminate its use by applying it to the explanation of prices and the prediction of returns, both of individual securities and market indices. We derive new closed form expressions for firm value; show that our model's explanation of stock prices is consistent with empirical evidence; and successfully forecast returns in two ways: in cross-section across the US stock market and in time-series for the S&P 500 index.

II. THE *P/B-ROE* VALUATION MODEL

The *P/B-ROE* model is a two-stage model of stock prices based on the change in price-to-book ratio (*P/B*) required to provide an exogenously determined expected shareholder return over a foreseeable horizon. It differs from most valuation models in that it does not explicitly discount future cash flows to arrive at a present price, but instead determines the future trajectory of price and valuation implied by the exogenously determined discount rate.

The *P/B-ROE* model is a forerunner of Estep's (1985) T-Model and Leibowitz's (1999) *P/E*-Orbit model, both of which are derived using similar logic. It differs from both in that its structure more readily invites empirical estimation of its parameters. A more complete discussion of its relationship to several well-regarded valuation models, along with a detailed derivation, is deferred to the appendices. Under the following simplifying assumptions:

1. Constant expected growth in book value at a rate g over the time interval $[0,T]$,
2. Market-determined expected shareholder return k,
3. Constant expected return on equity r during the interval $[0,T]$, and
4. Constant expected dividend yield d on book value during the interval $[0,T]$,

current price-to-book ratio, P/B_0, is related to the price-to-book ratio at time T, denoted P/B_T, via:

$$P/B_0 = P/B_T \times e^{(g-k)T} + \frac{d}{k-g}\left(1-e^{(g-k)T}\right). \quad (1)$$

The economic intuition that underlies the *P/B-ROE* model is simple, and particularly transparent for a non-dividend paying stock. If its expected return on equity is higher in the first stage than in the second, where it supports an equilibrium price/book ratio, its current *P/B* must be expected to decline, offsetting its high profitability so as to provide only the required shareholder return. On the other hand, if its expected return on equity is expected to be higher in the second stage, its *P/B* must be expected to rise until then to supplement its current depressed profitability and achieve the required shareholder return.

Equation (1) is non-linear, and has four independent parameters. This makes it most useful when good exogenous estimates of its inputs are available, as is likely the case in a corporate treasury, an investment bank, or within the research department of an asset management firm. It can, however, be well approximated and recast in a form that is better suited to statistical estimation. Exhibit 1 compares the valuation indicated by equation (1) for a firm that pays no dividends to one paying a dividend with a 6% yield on book value when k=8%, T=6 years, and P/B_6 = 2. The first valuation is log-linear with r and the second nearly so, especially over the domain of r that is of most practical interest. In Appendix II, we show that in the absence of dividends, equation (1) reduces to:

$$\ln(P/B_0) = \left[\ln(P/B_T) - kT\right] + rT\ . \quad (2)$$

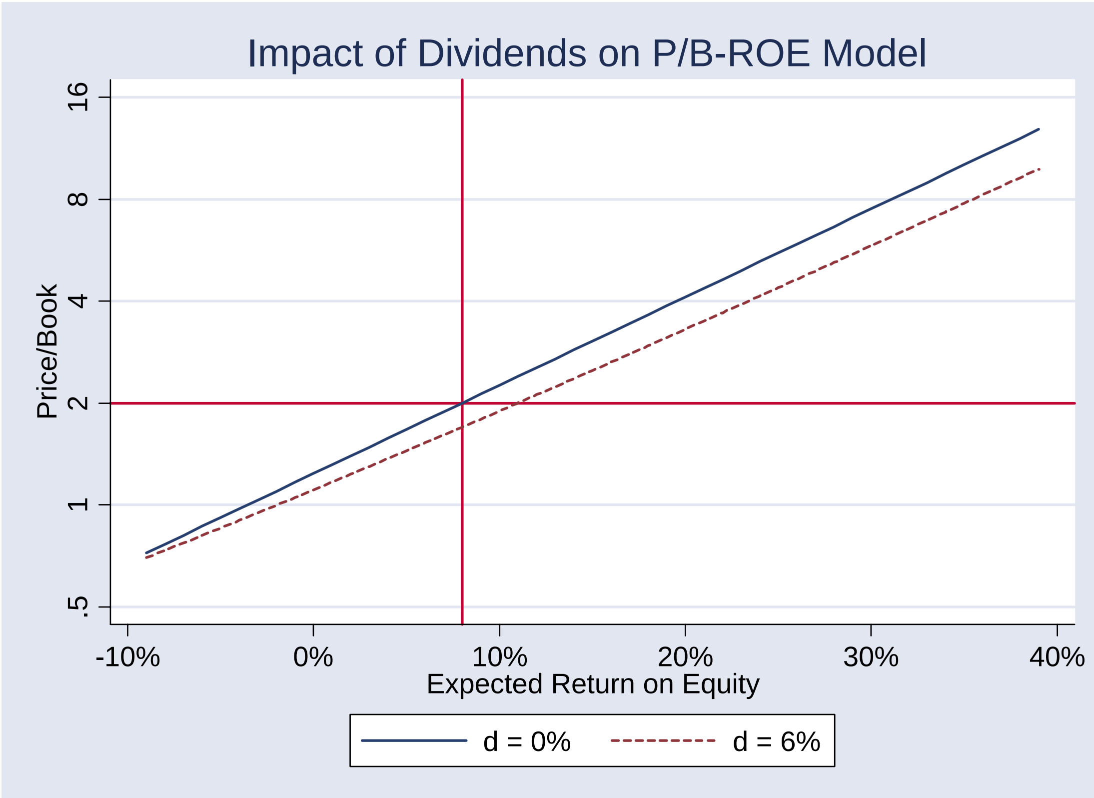

Exhibit 1. The Impact of Dividends on the *P*/*B*-*ROE* model

Equation (2) generalizes Wilcox's original *P/B-ROE* model, in which $P/B_T = 1$, causing $\ln(P/B_T)$ to vanish. Its utility for empirical measurement resides in three facts. First, it is surprisingly accurate for dividend paying firms. Second, *ROE*, unlike growth in earnings, tends to be persistent through time, providing an observable proxy for the market's expectation of *r*. (One may also attempt to improve on this proxy by incorporating analysts' estimates of future profitability, though de-biasing and rendering them exogenous is challenging). Third, equation (2) is linear in *r*, inviting efficient statistical estimation. We exploit these advantages in Section III to build cross-sectional explanatory and predictive models for individual securities, and in Section IV to build time-series explanatory and predictive models for the S&P 500 index.

If the market's expectation for future *r* is based largely on current *ROE*, then the slope of the empirical relationship linking P/B_0 to *ROE* gives us a measure of the foreseeable investment horizon *T*, while its intercept measures $\ln(P/B_T)+(r-ROE)T-kT$. If *ROE* is an unbiased estimate of *r*, the intercept measures $\ln(P/B_T)-kT$.

Note, however, that our estimate of *T* may be biased for several reasons, even if *ROE* is an unbiased estimate of *r*. First, random changes in book equity *B*, which appears in the denominator on both sides of the equation, tend to bias measured *T* upward, while random noise in the connection between *ROE* and *r* tends to bias it downward. In cross-section, it may blur across *T*'s that are more properly individual with each stock, and in time-series may be averaged rather than particular to the state of speculative interest. Nevertheless, being empirically estimated, it is in principle superior to the mostly subjective guesses made in quantitative applications of dividend discount models and their three-stage cousins.

III. CROSS-SECTIONAL COMPARISONS OF STOCK VALUES

In this section, we show how the *P*/*B*-*ROE* model can be used to explain cross-sectional differences in valuation and to predict future stock returns. Our analysis is based on a restricted subset of a large Value Line database that contains US stocks with at least five years of continuous history and whose fiscal year ends in December. The sample extends from 1984 to 2003 and is further restricted to company-years for which book values are positive, and for which *ROE* falls between -10% and +40%. The subset, whose sample size varies from about 700 companies in 1988 to about 1900 companies in 2003, includes both live and dead company files to minimize the impact of survivorship bias.

We first investigate the overall sensitivity of ln(*P/B*) to differences in *ROE*. To control for time variation in overall corporate profitability, we pool cross-sectional December data across years, and regress excess ln(*P/B*) against excess *ROE*, where both values are computed with respect to their cross-sectional medians. Log P/B increases by 3.66% for each 1% increase in *ROE*. This estimate, though subject to the biases earlier noted, indicates an average forecast horizon *T* for abnormal *ROE* of somewhat less than 4 years. It is an average across pooled data, and we should expect to find different sensitivities for groups of stocks differing in, among other ways, the degree to which current *ROE* mirror's the market's expectation for future *r*.

We next break our dataset into four quartiles based on the 5-year standard deviation of *ROE* and repeat our regression within each quartile. Quartile 1 contains companies with the most variation in *ROE* while quartile 4 contains companies with the least. Our estimates of ln(*P*/*B*)'s sensitivity to *ROE* in each quartile, along with the corresponding standard errors and t-statistics, are shown in Table 1. They increase monotonically from 2.37 years to 6.47 years as we go from the most volatile quartile to the least. Importantly, the explanatory power of the model increases as the volatility of *ROE* decreases, and as current *ROE* becomes a more reliable proxy for *r*.

	Quartile 1	*Quartile 2*	*Quartile 3*	*Quartile 4*	*Entire Set*
T	2.37	3.87	5.00	6.47	3.66
Std. Error (*T*)	0.082	0.077	0.085	0.094	0.042
t-stat	28.76	50.11	58.92	68.59	87.04
Adj. R-Squared	13%	32%	39%	47%	26%

Table 1. Investment Horizon vs. ROE Stability: 1988-2003

We report naïve t-statistics and standard errors in Table 1 with the knowledge and caution that our regression does not meet standard ordinary least squares (OLS) assumptions. Beyond the issues of offsetting biases noted in the preceding section, we have the problems of unequal residual variances and of clustering of residuals within industries and across time. For these reasons, adjusted R-squared should not be given its usual strong interpretation.

It is not possible to indicate the extent to which errors in explanatory variables incurred by using current *ROE* as a proxy for *r* are an issue without independent knowledge of the process by which expectations are formed. Consequently, we do not know the extent to which variability in *ROE* induces a downward bias in our estimate of *T*. In spite of this, it is useful to know that the estimated investment horizon is an increasing function of the stability of profitability.

Our dataset can be partitioned in many ways – grouping companies by the volatility of their *ROE* is just one possible partition. In particular, the regression can be run within an industry or peer group to obtain an estimate of *T* that can be used by corporate managers and financial analysts to determine the time horizon over which an investment is likely to earn an abnormal return, given the dynamics of that industry. We know of no other empirical method that so readily yields an estimate of *T*, a parameter that is central to both corporate decision-making and the construction of sophisticated valuation models.

We next examine the stability of the investment horizon *T* across time. Table 2 tabulates our estimate of *T*, along with the median *ROE*, in each year for the full sample OLS regressions linking ln(*P/B*) to *ROE*. The estimated value of *T* ranges between 2.55 and 4.54 years. In addition, the explanatory power of our model remains stable over time, as does its standard error. Similarly consistent results, not shown, were obtained in each of the earlier-described quartiles.

Year	*T*	*Std. Err. (T)*	*Adjusted R^2*	*Median ROE*
1988	2.55	0.18	22%	14.4%
1989	2.97	0.19	21%	13.6%
1990	4.32	0.17	35%	12.1%
1991	4.18	0.18	30%	11.4%
1992	3.46	0.16	26%	11.3%
1993	3.01	0.15	22%	12.2%
1994	3.14	0.15	24%	12.5%
1995	3.11	0.15	22%	13.2%
1996	3.32	0.14	26%	13.3%
1997	3.60	0.13	32%	13.5%
1998	4.22	0.16	31%	12.9%
1999	4.25	0.18	27%	13.1%
2000	4.54	0.18	29%	13.4%
2001	3.57	0.16	24%	10.2%
2002	3.52	0.14	28%	10.6%

Table 2. Estimates of *T* by Year: 1988-2002

Extracting information from the intercepts of our regressions is complicated because information about required shareholder return *k* is confounded with variation in the time-series of expectations for P / B_T and also with any biases inherent in using *ROE* as a proxy for *r*. Disentangling these would be speculative at our current state of knowledge, and is left for further research.

Most readers will be interested in a valuation model's ability to predict future stock returns. This is distinct from stock price explanation, because if the insights of a valid explanatory model are *fully* impounded in market prices, it will not help predict future returns. On the other hand, if future returns are negatively correlated with discrepancies between market prices and a valuation model's fitted values, and if prices revert to the model's predicted values, it suggests that the model is a better measure of some true underlying economic reality than contemporaneously observable price data.

We therefore test the power of the *P*/*B-ROE* model's December residuals (i.e. the difference between each stock's actual *P*/*B* and the *P*/*B-ROE* model's fitted *P*/*B*) to predict stock returns in the following year. We are acutely aware that our results may not be exploitable in the first several months, because earnings for the preceding year have not yet been reported or even estimated with precision. However, predictability persists for up to nine months, and well past any reasonable estimate of reporting lag. The correlation between year-end *P/B-ROE* residuals and excess returns (i.e. in excess of the sample median return) for each month, as well as for three-month periods ending in March, June and September, are shown in Table 3. The monthly correlation is strongest in January and then declines over the next nine months.

Period	*Correlation With Year-end P/B-ROE Residual*
January	-0.160
February	-0.059
March	-0.041
April	-0.031
May	-0.032
June	-0.038
July	-0.043
August	-0.004
September	-0.001
October	0.047
November	-0.002
December	-0.019
January-March	-0.157
April-June	-0.058
July-September	-0.026

Table 3. Correlation Coefficients: December Residuals vs. Future Returns. All Years Combined

The results of OLS regressions relating return in three disjoint three-month periods to the end of prior-year valuation residual are shown in Table 4. Negative residuals forecast positive excess returns, and do so well beyond any reasonable allowance for reporting lags. This is of particular interest since fourth quarter earnings surprises are the only incremental information not already impounded in prices as the result of quarterly 10-Q filings, press releases and analysts' estimate changes available earlier in the year.

We leave open the question of whether or not these predictable excess returns represent risk premia or the closure of market inefficiencies. We confess to some skepticism as to whether the Fama-French (1995) book-to-market premium is entirely responsible for our results, both because many firms with high *P*/*B* ratios but even higher profitability have negative residuals, and because the rate of decay of predictability seems too rapid for the erosion of a risk premium.

Period	*OLS Coefficient*	*Std. Error*	*t-statistic*	*Adjusted* R^2
January-March	-0.055	0.0026	-21.19	2.22%
April-June	-0.022	0.0025	-8.69	0.38%
July-September	-0.011	0.0024	-4.50	0.10%

Table 4. Regression coefficients: December Residuals vs. Future Returns. All years combined.

What of consistency of return predictability? As value-oriented investors have often found, a long-term statistical edge in forecasting excess returns is often accompanied by periods of underperformance. As can be seen from Table 5, *P*/*B-ROE* residuals have the same property. They were enormously predictive in the aftermath of the recent bubble, as shown by a t-statistic of -12.81. (Incidentally, the simultaneous decline in the market might suggest that they do not derive their predictive power through a risk premium.) In 6 of the 15 years studied, however, excess returns over benchmarks would have been moderately negative. In the interest of brevity, we present pooled results only for the second quarter of each year. Those for the first and third quarters are qualitatively similar, but are stronger and weaker respectively.

Year	*Coefficient*	*Std. Error*	*t-Statistic*
1989	0.006	0.011	0.49
1990	0.047	0.010	4.84
1991	-0.045	0.009	-4.77
1992	-0.036	0.009	-4.25
1993	0.024	0.010	2.52
1994	-0.059	0.008	-7.60
1995	-0.003	0.011	-0.24
1996	0.012	0.009	1.46
1997	0.013	0.009	1.39
1998	-0.026	0.009	-2.81
1999	-0.021	0.011	-1.95
2000	0.021	0.010	2.18
2001	-0.016	0.008	-1.92
2002	-0.115	0.009	-12.81
2003	-0.063	0.012	-5.17

Table 5. April-June Regression Coefficients Against Prior Year-End Residuals, 1989-2003

IV. A TIME-SERIES MODEL FOR THE US MARKET

Explaining and predicting the price of a single stock based on its history is difficult, as price can depend on many hard-to-quantify inputs, such as the competitive dynamics of an industry or even the outcome of a lawsuit. These modeling problems do not put insuperable hurdles in front of the CEO or CFO trying to understand how best to increase the corporation's value, but they do put the outside observer at a distinct disadvantage. For an investor, forecasting the stock returns of individual stocks using a time series *P/B-ROE* model poses great challenges, and we refrain from attempting to do so here.

Modeling the market as a whole, however, is easier than modeling individual stocks, as aggregation diversifies away non-systematic risk and creates a composite whose prices are more readily explained by a single-stage or two-stage model. Even so, the task is not trivial, as various elements of the model exhibit significant time variation, and we have relatively few observations from which to estimate model parameters.

There is a rich literature, exemplified by Ilmanen (2003) and Philips (1999, 2003), on forecasting k using valuation ratios such as *P/B* and measures of profitability similar to r. While these models appear to have good explanatory power for long-term equity returns, they have little or no power to forecast short-term market fluctuations. In this section, we show that the *P/B-ROE* model has explanatory power for the time-series of market prices and that its residuals have had useful forecast power for short to intermediate-term market returns.

We start our study by applying a variant of equation (2) to monthly values of the Standard & Poor's 500 index (S&P 500). We proxy r with current reported GAAP return on equity for the S&P 500, and k with Moody's seasoned Baa bond interest rate, splitting this nominal rate into real and inflationary parts. While the yield on corporate bonds is clearly a downward-biased estimate of the expected return of stocks, since it does not include the full equity risk premium, its fluctuations do have the advantage of being more *exogenous* with respect to the dependent variable than estimates derived from the history of stock prices and estimates of earnings and economic growth. It is also useful to account separately for current inflation, as it both amplifies required shareholder return k and positively biases current *ROE* with respect to r. We measure inflation with the 12-month growth in the consumer price index (CPI).

All S&P 500 data is sourced from Barra Inc., distributed on their website by agreement with Standard & Poor's, and from Morgan Stanley Quantitative Strategies (Institutional Equity Division), which independently computes the aggregated fundamental characteristics of the index over a longer period. Interest and inflation rates are sourced from the Federal Reserve Bank of St. Louis' FRED II database. The history of our model's inputs is shown in Exhibit 2. The chart gives us an immediate insight into the bull market of the late 1990's – *ROE* stayed near long-term highs while both inflation rates and real interest rates declined.

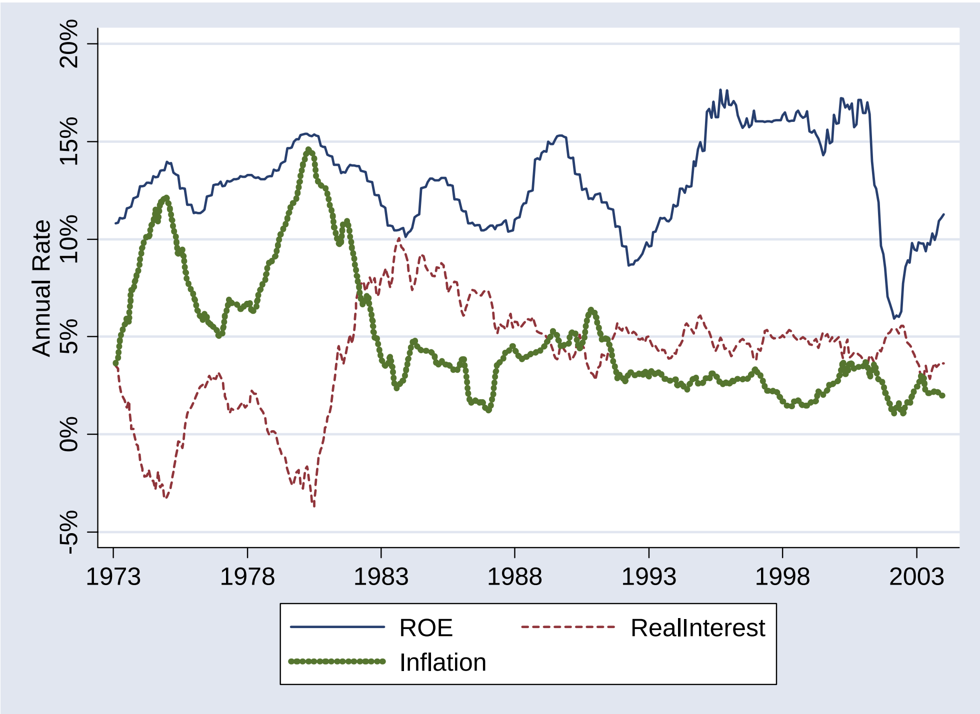

Exhibit 2: History of Inputs for S&P 500 *P*/*B*-*ROE* Model

We explain the natural log of the S&P 500's price-to-book ratio by a linear combination of its reported *ROE*, 12 month trailing CPI, and the difference between Moody's seasoned Baa bond interest rate and 12 month trailing CPI. Although our model cannot be validated *internally* because of omitted variables that create strongly autocorrelated residuals, it can be validated *externally* by testing the ability of its no-look-ahead residuals, calculated from successive models using an expanding window, to predict the subsequent returns of the S&P 500 index. Our regressions are of the form

$$\ln(P/B_0) = a + b \times ROE + c \times \text{RealInterest} + d \times \text{Inflation}\,, \qquad (3)$$

and the results of an explanatory regression over the entire 373 month period are shown in Table 6. These results are of little statistical value because the residuals are highly autocorrelated ($\rho > 0.95$). And, of course, as investors during the period, much of the data would not have been available to us. However, the results are of considerable interest if one holds the *PB-ROE* model as a Bayesian prior, because they are quite consistent with that model.

	Coefficient	*Std. Error*
ROE	6.287	0.501
Inflation	-15.864	0.490
RealInterest	-8.004	0.541
Constant	1.083	0.084
Adjusted R^2	76%	

Table 6. Full Period P/B-ROE model for the S&P 500

We next remove look-ahead bias by performing a series of expanding window regressions, expanding the sample of observations by one month as each month passes, with an initial five-year warm-up period to settle the model coefficients. Exhibit 3 compares actual P/B to the fit of the expanding window regressions. Valuation multiples continued to rise somewhat faster than our model's estimate from the mid 1980s till 2001, except for a brief moment immediately after the crash of 1987. Note the correspondence in Exhibit 3 between the shorter-term wiggles in the explanation and the actual movement of the S&P 500. There is even an exaggerated prediction of decline and recovery in 2002: this may be due to the extensive write-downs of accumulated past profit reporting sins that took place at that time. As with the preceding model, the autocorrelation of residuals is over 0.9, impairing the model's direct statistical validation. In spite of this, the model's residuals have significant forecast power for future returns.

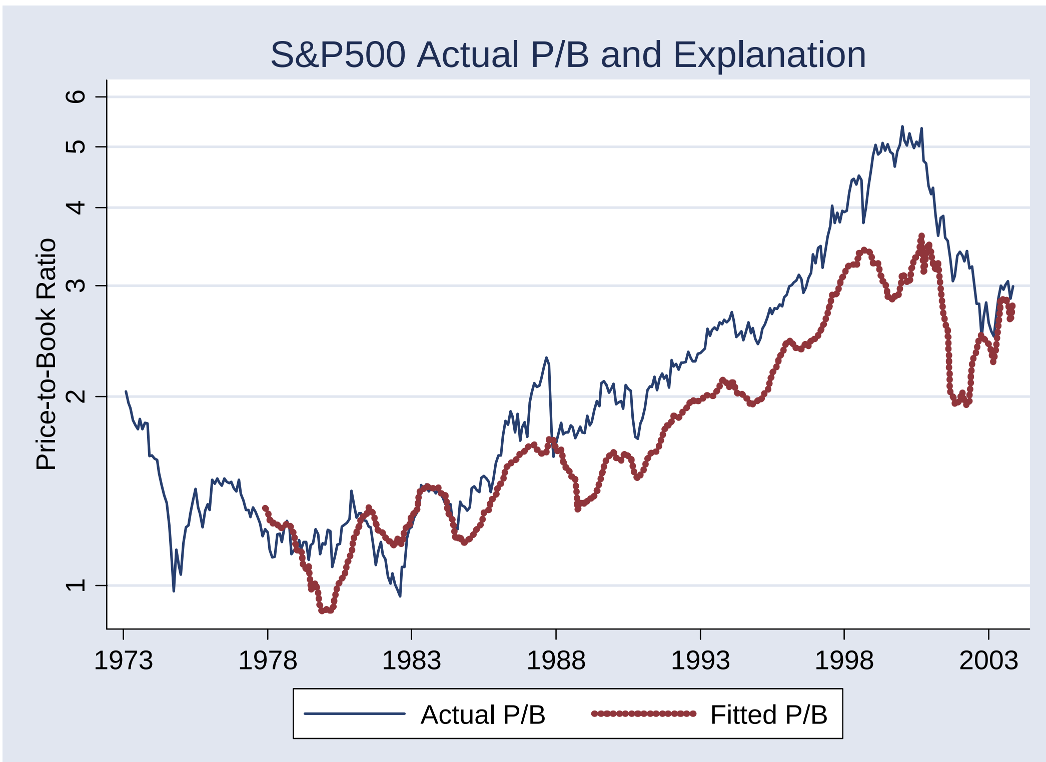

Exhibit 3: P/B-ROE model. Fitted and Actual P/B ratios

Exhibit 4 shows the cross-correlogram of 1 month S&P 500 returns and the expanding window *PB-ROE* regression residuals at different monthly lags and leads. The negative correlations on the right show that residuals are negatively correlated with subsequent returns for many months into the future. It is also interesting that model residuals, which can be affected either by stock price changes or by changing fundamentals, are led by returns for a few months prior to measurement. This is indicated by positive correlations for negative lags.

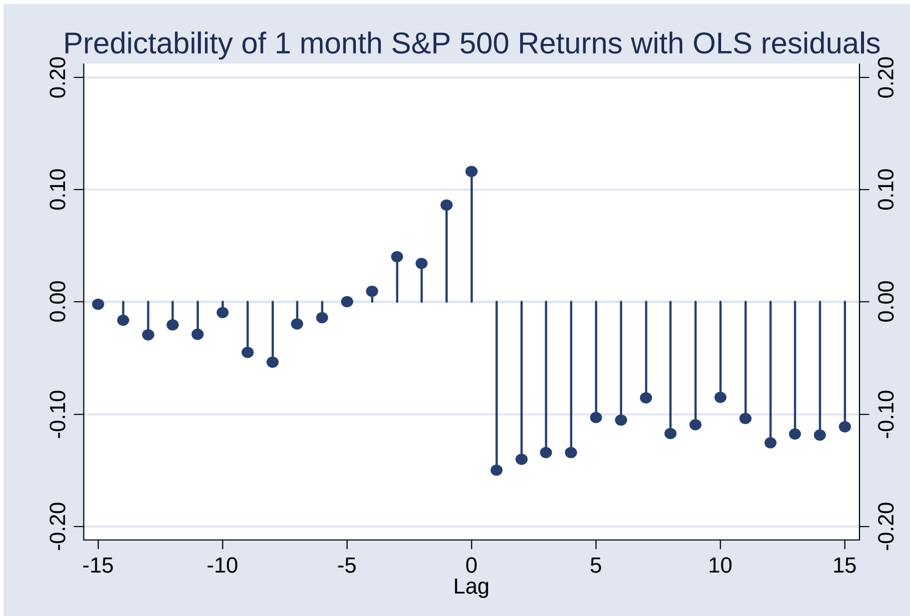

Exhibit 4: Cross Correlation – Model Error vs. Future 1 Month S&P 500 Returns

Table 7 shows the results of regressing the returns for the next 1, 3 and 6 months against the previous month's *P/B-ROE* residual. All standard errors are computed using the Newey-West (1987) correction for overlapping observations, and all coefficients are significant at the 2% level. The table shows just how powerful the relationship is, even using our crude proxies for inflation and *k*, and it is clear that such a relationship would have been of interest to investors engaging in Tactical Asset Allocation (TAA)

	Coefficient	*Std. Error*	*t*	*P>|t|*	*95% Confidence Interval*	
1 month return	-0.0418	0.017	-2.41	0.016	-0.0759	-0.0077
3 month return	-0.1199	0.050	-2.39	0.018	-0.2188	-0.0210
6 month return	-0.2243	0.093	-2.41	0.017	-0.4077	-0.0409

Table 7: Predictive Power of S&P 500 P/B-ROE residuals

V. CONCLUSION

We have described and tested a model of security prices that is both simple and analytically tractable, both economically and statistically significant, and which complements more widely-used valuation-based approaches. Its advantages lie in its elegance and simplicity, its widespread applicability, both in cross-section and in time-series, and its amenability to empirical estimation. It is useful based on widely available reported data, and its power may well be further enhanced by the use of improved proxies for expected return on equity, more accurate estimators of required shareholder return, and by restricting its use to particular subsets of the market where these proxies are most accurate.

VI. ACKNOWLEDGEMENTS

Eric Lufkin of Morgan Stanley Quantitative Strategies (Institutional Equity Division) graciously provided us with data on the S&P 500. Daniel Rie and Evan Schulman made numerous helpful suggestions. It is a pleasure to acknowledge their contributions. All remaining errors are our own.

APPENDIX I. COMPARISON TO OTHER MODELING APPROACHES

The PB-ROE model has three features that distinguish it from most approaches discussed in the investment literature. First, it is a two-stage model that explicitly models the time horizon *T* over which competition will erode a firm's ability to earn an abnormal *ROE*. It is therefore inherently *more realistic* than a single-stage model, though without the significant increase in complexity that invariably accompanies a three-stage model.

Second, because the PB-ROE model is structured around return on equity and dividend yield on book, both of which (unlike growth in earnings) are often highly persistent, it allows the use of both forward looking estimates and historical measurements. Consequently, it is inherently *more objective* than most valuation models.

Third, because of the simple structural form of equation (2) and the suitability of reported *ROE* as an expectational proxy, the *PB-ROE* model can take advantage of large collections of comparative data through statistical estimation. Even though it contains a number of approximations, its amenability to empirical estimation from data makes it inherently *more informative* than models focused on a single company in isolation.

These three advantages of realism, objectivity and information extraction are evident in varying degrees when the *PB-ROE* model is compared to other models. The Gordon-Shapiro growth model (1962) is a single stage dividend discount model that approximates dividends as growing exponentially at a constant rate in perpetuity. It has found some success at the level of the aggregate market, but is not robust when applied to individual securities, which may have expectations for substantial temporary abnormal growth and dividend policy. Also, its form suggests that the only way to maximize shareholder value is to maximize dividend growth, without specifying how this should be done.

The Residual Income Valuation Model of Ohlson et. al. (1995, 2000), also known as the Edwards-Bell-Ohslon Ohlson equation, is a period-by-period restatement of the dividend discount model in earnings garb, and takes the form:

$$P_t = B_t + \sum_{i=1}^{\infty} \frac{E[(r_{t+i} - k) \times B_{t+i-1}]}{(1+k)^i} .$$

It has appropriately attracted a great deal of attention in recent years, and has been used both to value individual securities and the market (as in Herzberg (1998) and Lee, Myers and Swaminathan (1999)) and to estimate the expected return of the market (as in Philips (1999, 2003) and Claus and Thomas (2001)).

It immediately yields the right approach to enhancing shareholder value – maximize the spread between the return on capital and the cost of capital – and readily allows the integration of economic insights such as the convergence over time of the return on capital to the cost of capital. Also, it focuses on return on equity, which is more forecastable from its own past history than is growth in earnings or dividends. However, like Gordon growth model, it gives us no guidance on how far into the future we should estimate abnormal return on equity, and its structure is not linear. Herzberg (1998) shows that its conceptual advantages do not assure the EBO model exceptional predictive power at the level of individual securities.

Leibowitz and Kogelman's (1994) Franchise Factor Model and Leibowitz's (2000) Spread-Driven Dividend Discount Model both structure a firm as a combination of two entities, one containing all the current investments of the firm and the other containing its future investments. The *ROE* of the firm's future investments is not necessarily equal to that of its existing investments. This is a useful distinction, since a corporate insider can meaningfully construct separate expectations for each. Both models relate the *P/E* of the firm to that of the market and the expected *ROE* of its new investments. However, to the best of our knowledge, they have not been empirically tested, and so appear most suitable for use when exogenous estimates of the required inputs are readily available.

Estep's (1985) T-Model is derived using the same logic as in Wilcox (1984), but with an exogenously determined investment horizon, which he fixes at 20 years. No attempt is made to estimate the horizon from data. The T-model decomposes total return in terms of growth, *ROE* and change in *P/B* over this horizon. Estep then uses the expression for total return as a proxy for *k*, and shows that his model has some forecast power for future returns.

Leibowitz (1999), too, uses logic similar to Wilcox (1984), but focuses on a stock's *P/E* ratio, and shows that it must evolve along certain trajectories (or orbits) if its expected return is exogenously determined. He relates the shape of these orbits to initial and final growth and *k*, and illustrates a number of errors that analysts make. In particular, he shows that a popular valuation technique – projecting growth over some horizon and assuming no change in *P/E* – is fundamentally flawed, as it implies a change in *k* over the first phase. Leibowitz's article is normative – it offers a prescription for how investors ought to think about expected and realized returns in a two-stage model – but he does not empirically test its predictive power.

APPENDIX II. MODEL DERIVATION

Notation:

P/B_t = Price-to-Book ratio at time *t*,

T = Time horizon over which the firm earns an abnormal *ROE*,

g_t = Expected rate of growth of book value at time *t*, assumed constant *g* for $t \le T$, and constant g_{eq} for $t>T$,

D_t = Cumulative dividend process at time *t*

d_t = Expected ratio of the instantaneous rate of dividend payments to book value at time *t*, assumed constant *d* for $t \le T$, and constant d_{eq} for $t>T$,

$r_t = g_t + d_t$ = Expected instantaneous return on equity at time *t*, assumed constant *r* for $t<T$, and constant r_{eq} for $t>T$, and

k_t = Required expected shareholder return at time *t*, assumed constant for all $t>0$.

By definition:

$$k_t = \frac{\frac{\partial P_t}{\partial t} + \frac{\partial D_t}{\partial t}}{P_t} = \frac{1}{P_t}\frac{\partial P_t}{\partial t} + \frac{1}{P_t}\frac{\partial D_t}{\partial t}, \tag{A.1}$$

and

$$P_t = B_t \times P/B_t . \tag{A.2}$$

Differentiate (A.2) with respect to time and divide both sides by P_t to give

$$\frac{1}{P_t}\frac{\partial P_t}{\partial t} = \frac{1}{P/B_t}\frac{\partial P/B_t}{\partial t} + \frac{1}{B_t}\frac{\partial B_t}{\partial t} . \tag{A.3}$$

Combining (A.1) and (A.3) we get

$$k_t = \frac{1}{P/B_t}\frac{\partial P/B_t}{\partial t} + \frac{1}{B_t}\frac{\partial B_t}{\partial t} + \frac{1}{P/B_t}\frac{1}{B_t}\frac{\partial D_t}{\partial t} . \tag{A.4}$$

Using the definition of dividend yield on book and growth in book value

$$k_t = \frac{1}{P/B_t}\frac{\partial P/B_t}{\partial t} + g_t + \frac{1}{P/B_t}d_t = g_t + \frac{1}{P/B_t}\left[\frac{\partial P/B_t}{\partial t} + d_t\right], \tag{A.5}$$

which can be rearranged to give

$$P/B_t \times (k_t - g_t) = \frac{\partial P/B_t}{\partial t} + d_t . \tag{A.6}$$

This is a first-order homogenous ordinary differential equation, and is easily solved if *k*, *g*, and *d* are constant, as we have assumed. In this special case, (A.6) reduces to

$$\frac{\partial P/B_t}{\partial t} = P/B_t \times (k - g) - d . \tag{A.7}$$

If P/B_t is to be stable, as it is in the Gordon-Shapiro growth model, the left-hand side of Equation A.7 must be zero, and

$$P/B_t = P/B_0 = \frac{d_{eq}}{k - g_{eq}} . \tag{A.8}$$

If the clean surplus assumption holds, and if there is no change in capital structure, dividends are the difference between earnings and retained earnings, and the growth in book value is exactly equal to the amount of retained earnings, so that *d* = *r*-*g*, and

$$P/B_t = P/B_0 = \frac{r_{eq} - g_{eq}}{k - g_{eq}}. \tag{A.9}$$

Returning to the more realistic two stage model allowing abnormal growth and dividends until time *T*, we want to solve the differential equation (A.7) with the boundary condition P/B_T $= \frac{d_{eq}}{k-g_{eq}} = \frac{r_{eq}-g_{eq}}{k-g_{eq}}$. The general form of the solution is

$$P/B_t = P/B_0 \times e^{(k-g)t} + \frac{d}{k-g} \times \left(1 - e^{(k-g)t}\right). \tag{A.10}$$

When we enforce the boundary condition at *T* we get

$$P/B_0 = P/B_T \times e^{(g-k)T} + \frac{d}{k-g}\left(1 - e^{(g-k)T}\right). \tag{A.11}$$

But $\frac{d}{k-g} = \frac{r-g}{k-g} = P/B*$, the *P/B* that would prevail if the initial growth rate and ROE persisted in perpetuity, i.e., P/B_0 is a weighted sum of P/B_T and $P/B*$, with the latter having the possibility of a negative weight. If $d = 0$, $g = r$ and (A.11) reduces to

$$\ln(P/B_0) = \ln(P/B_T) + (r-k)T, \tag{A.12}$$

which lends itself to econometric measurement using linear regression if we have proxies for P/B_T, *r* and *k*. Wilcox (1984) postulates that after *T*, r_{eq} will equal *k*, forcing P/B_T to equal 1. This assumption has the great benefit of making ln(*P/B*) an approximately linear function of *r* if *d* is not too large. However, the convergence of r_{eq} to *k* is not assured, and in general we have:

$$P/B_T = \frac{r_{eq} - g_{eq}}{k - g_{eq}}. \tag{A.13}$$

On substituting (A.13) into (A.11), we see that

$$P/B_0 = \frac{r_{eq} - k}{k - g_{eq}} e^{(g-k)T} + \frac{r-g}{k-g} - \frac{r-k}{k-g} e^{(g-k)T}. \tag{A.14}$$

But $\frac{r-g}{k-g} = P/B*$, the price-to-book ratio that would prevail if the initial growth rate and *ROE* persisted in perpetuity, so that (A.14) can be rewritten as

$$P/B_0 - P/B* = \left(\frac{r_{eq} - k}{k - g_{eq}} - \frac{r-k}{k-g} \right) e^{(g-k)T}. \tag{A.15}$$

As $e^{(g-k)T} > 0$, both $(P/B_0 - P/B*)$ and $\left(\frac{r_{eq} - k}{k - g_{eq}} - \frac{r-k}{k-g} \right)$ are of the same sign, and their ratio is positive. Consequently, we can take natural logarithms to give

$$\ln \left(\frac{P/B_0 - P/B*}{\frac{r_{eq} - k}{k - g_{eq}} - \frac{r-k}{k-g}} \right) = (g-k)T. \tag{A.16}$$

Unfortunately, (A.16) is not easily estimated. We therefore focus our efforts on (A.12). If *d* is small, as it has been for most stocks in recent times, (A.12) is a good approximation to (A.11), though it somewhat underestimates *T* and the *P/B* of stocks with high *d* and low *r*, and overestimates the *P/B* of stocks with high *d* and high *r*.

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